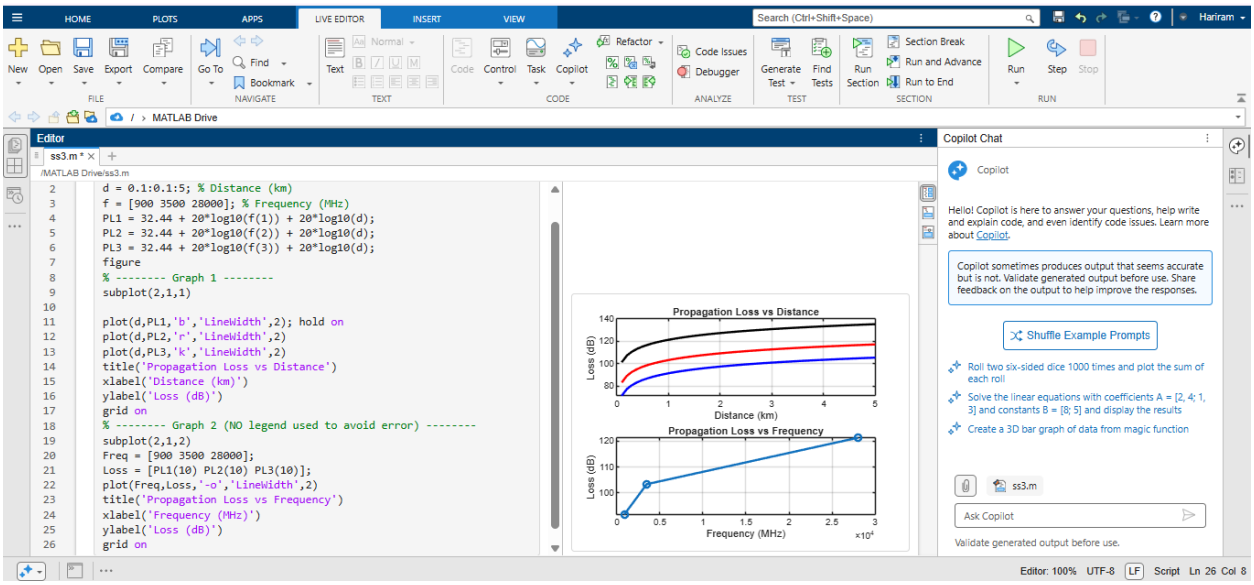
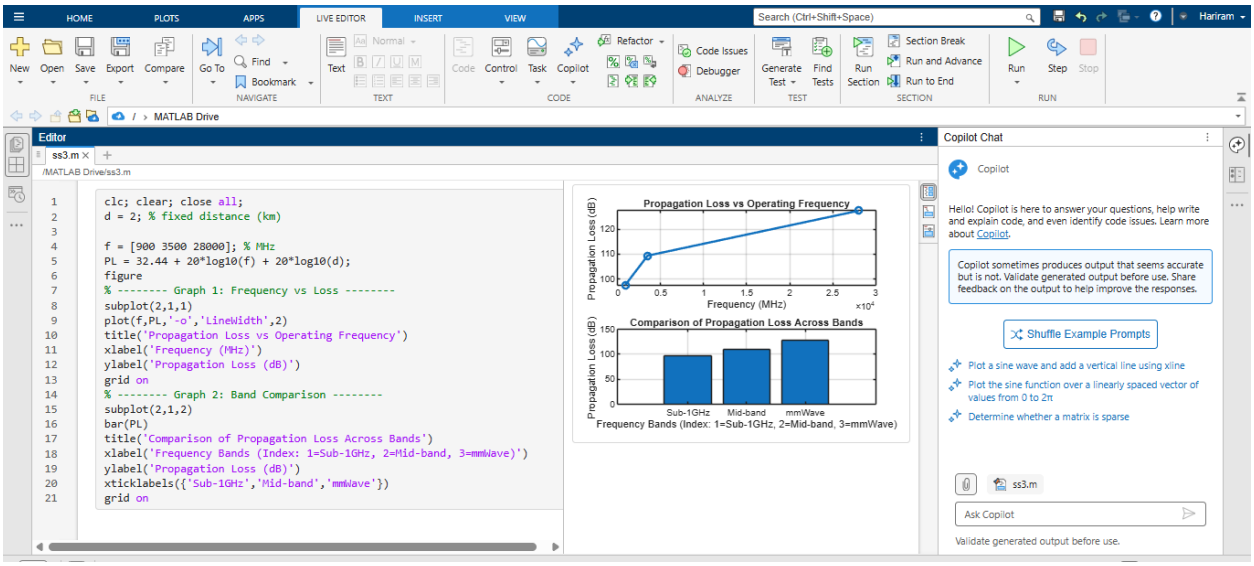
EXP 3(i)



```
d = 0.1:0.1:5; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% -------- Graph 1 --------
subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
% -------- Graph 2 (NO legend used to avoid error) --------
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```

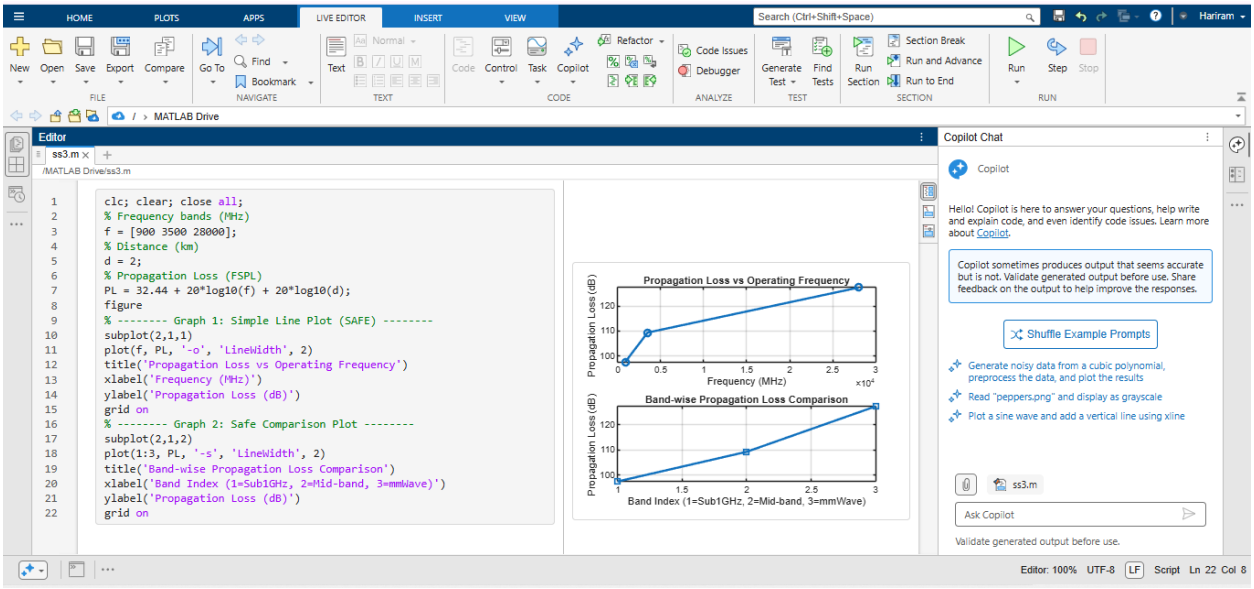
(ii)



```
clc; clear; close all;
d = 2; % fixed distance (km)

f = [900 3500 28000]; % MHz
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Frequency vs Loss --------
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Band Comparison --------
subplot(2,1,2)
bar(PL)
title('Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
grid on
```

(iii)


HOME
PLOTS
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LIVE EDITOR
INSERT
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Search (Ctrl+Shift+Space)
Hariram
New
Open
Save
Export
Compare
Go To
Find
Bookmark
Text
Code
Control
Task
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
FILE
NAVIGATE
TEXT
CODE
ANALYZE
TEST
SECTION
RUN
MATLAB Drive
Editor
ss3.m
/MATLAB Drive/ss3.m
clc; clear; close all;
% Frequency bands (MHz)
f = [900 3500 28000];
% Distance (km)
d = 2;
% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Simple Line Plot (SAFE) --------
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Safe Comparison Plot --------
subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on
Propagation Loss vs Operating Frequency
Propagation Loss (dB)
Frequency (MHz)
Band-wise Propagation Loss Comparison
Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)
Copilot Chat
Copilot
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Shuffle Example Prompts
Generate noisy data from a cubic polynomial, preprocess the data, and plot the results
Read "peppers.png" and display as grayscale
Plot a sine wave and add a vertical line using xline
ss3.m
Ask Copilot
Validate generated output before use.
Editor: 100%
UTF-8
LF
Script
Ln 22 Col 8